Any Old Theory Will Do: Why Cause-and-Effect Performance Links Form Parsimonious Mental Models of Complex Strategic Environments

"He who loves practice without theory is like the sailor who boards a ship without a rudder."
~Leonardo da Vinci

ABSTRACT

Mental models can help decision-makers understand complex strategic environments and thus choose better strategies. Recent research in the theory-based view of strategy argues that theories– mental models that represent cause-and-effect relationships between strategic choices– are especially useful in strategy selection because they effectively guide experimentation and learning. In this paper, I use a computational model to explain why the cause-and-effect links of theories outperform the associative links of the mental models prevalent in strategic cognition. I show that, by causally ordering choices, theories improve how efficiently decision-makers use performance feedback to search for strategies. My work suggests that even if causal models are less accurate representations of the strategic environment than associative models, their simplified and directed representation consistently leads decision-makers to more successful strategies.

KEYWORDS: Theory-Based View, Mental Models, Strategic Cognition, Decision-Making, NK Model

1. INTRODUCTION

The theory-based view of strategy proposes that theories – mental models that specifically represent cause-and-effect relationships between strategic choices — find successful strategies (Felin et al. 2024, Camuffo et al., 2024). This theory-based view argues that specifying a causal pathway linking strategic choices to performance helps decision-makers more effectively experiment and learn, allowing them to select more successful strategies over time (Camuffo et al. 2024, Valentine et al. 2024, Sorenson 2024, Ehrig and Schmidt 2022). Yet, while there is some evidence that theories can improve strategic search (Camuffo et al. 2020), there is no model that explains why these mental models with cause-and-effect performance links would find better strategies than the mental models used in the large body of work that preceded this theory-based turn.

Following the work of Porac, Thomas, and Baden-Fuller (1989), a stream of cognitive strategy research has long argued that mental models help decision-makers find more successful strategies (see Kaplan 2011 for a review). One of the core mechanisms through which these mental models improve strategic search is through simplifying the complex strategic environment (Gavetti and Levinthal 2000, Bingham and Eisenhardt 2011). By focusing on a subset of strategic choices, decision-makers can more easily search the environment, better attribute performance variation, and avoid getting stuck on poor strategies (Csaszar and Levinthal 2016).

The major difference between the mental models as conventionally conceptualized in cognitive strategy research and the mental models in the theory-based view is the type of performance relationship they represent between strategic choices. In the cognitive strategy research tradition, mental models have implicitly associative performance links. These mental models, which I will refer to as associative mental models, represent the performance of two strategic choices as related (even causally related), but do not specify the direction of the causality (see Gavetti and Levinthal 2000, Csaszar and Levinthal 2016,

Csaszar and Ostler 2020).[1] In the theory-based view of strategy, mental models have cause-and-effect performance links. These mental models, which I will refer to as causal mental models or theories, represent the selection of one strategic choice as the performance antecedent of another (see Camuffo et al. 2024, Sorenson 2024).

This difference in performance link type of mental models is likely to impact the performance of strategies selected by them because it affects how accurately and simply the mental model represents the strategic environment. Previous research suggests that both simple and accurate mental models find high-performing strategies, because simple models are easier to understand and accurate models are right (Gavetti and Levinthal 2000, Gary and Wood 2011). When the strategic environment is simple, mental models can be both simple and accurate. However, as complexity in the strategic environment increases, decision-makers using mental models to find strategies face a trade-off between simply and accurately representing it. In these complex environments, cognitive strategy research shows that associative mental models improve strategy selection through representing fewer strategic choices (Csaszar and Levinthal 2016).

While this prior work suggests that simply representing complex strategic environments is likely to generate more performant strategies, it tends to explore one facet of simplification, that of considering fewer strategic choices (see Martignoni et al. 2016 for an exception). However, mental models are made up of two components: a set of strategic choices and a set of performance relationships between those choices. Thus, another way a mental model could simplify the strategic environment is through representing fewer performance links between strategic choices, which is what the cause-and-effect performance links of theories do.

[1] Contingency functions ($f(x)$) which specify the interdependence of two variables (x_1 and x_2) as an interaction term (x_1x_2) such that $f(x) = \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$, can always be differentiated in terms of both variables in the interaction ($\frac{\partial f}{\partial x_1} = \beta_1 + \beta_3 x_2$ and $\frac{\partial f}{\partial x_2} = \beta_2 + \beta_3 x_1$). Thus, these functions, by allowing both variables in the interaction to vary as a function of the other, do not specify causal direction of dependence between them (for example Csaszar and Ostler 2020).

I propose that theories simplified and directed representation of performance relationships enables decision-makers to use performance feedback more effectively during strategic search. Specifically, by representing fewer performance dependencies, theories allow decision-makers to use more performance feedback to make each strategic choice. Moreover, because theories order strategic choices into a causal sequence, decision-makers use more performance feedback to make choices earlier in the sequence, aligning feedback use with perceived strategic importance. Together, this greater quantity and better allocation of performance feedback improve decision-makers' understanding of performance, which ultimately leads to more performant strategic search.

Using a model that compares how decision-makers employ different mental models to search for strategies in increasingly interdependent environments, I find that decision-makers who use theories to direct their search for strategies better understand what drives performance in the strategic environment over time, ultimately finding different and better strategies than associative mental models. This is true even in highly interdependent environments where associative mental models are more accurate representations of the strategic environment, showing that the simple representation of performance dependence in theories is a parsimonious way for decision-makers to select better strategies.

My work formalizes the core claim of the theory-based view, mainly, that holding a theory improves strategic search and organizational performance. It does so by developing a conceptual framework of how the difference in performance link type between theories and mental models as typically operationalized in strategy research, leads to differences in the way decision-makers use performance feedback and attribute performance to strategic choices. Particularly, by showing that the causally directed links of theories allow decision-makers to use more performance feedback in a way that is aligned with perceived strategic importance, my findings provide an analytical explanation for the performance benefits of using theories over more accurate associative mental models when searching for strategies. This finding challenges some prevailing notions of the use of causal reasoning for decision-making, which has suggested that if causality is useful at all, it is for discursive reasons and not

comprehension (Mercier and Sperber 2017, Sperber et al. 1995). Overall, I suggest that theories are a powerful tool for organizational decision-makers looking to form successful strategies.

2. CHOOSING A STRATEGY WITH DIFFERENT MENTAL MODELS: AN EXAMPLE

To illustrate the difference between strategic decision-making using a theory versus an associative mental model, I use the example of a founder of a health tech startup trying to decide on three key strategic choices in her startup, where she needs to select two features of her AI model built to detect early signs of cancer (data type and learning type), and also needs to choose who to hire to build the AI model (hiring choice). The key difference in decision-making is that theories prompt sequential decisions, while associative mental models prompt a set of simultaneous strategic choices.

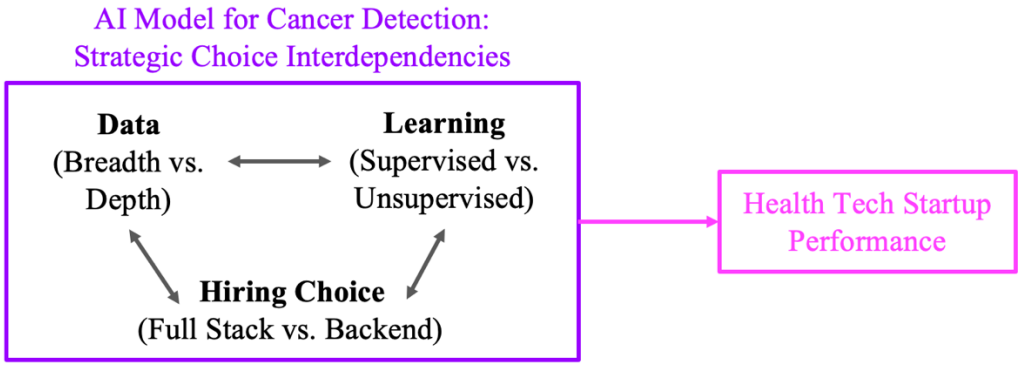
In order to build a high-performance startup around an AI model that detects cancer, the startup founder has many interdependent choices to consider. For example, the founder needs to decide which data her AI model should be trained on, where she could either choose a dataset with a lot of breadth (a few indicators for many different types of cancer) or depth (many indicators for a single cancer type, i.e. breast cancer). A model that uses a wide breadth of data will make the startup more marketable to a wider set of potential customers, while a model that has depth of data is likely to make more accurate predictions for a particular type of cancer, making it more valuable to a smaller set of customers.

However, the impact of the choice of data on the performance of the start-up is also dependent on the other model features that the founder selects, such as the way the model learns. In selecting the way the AI model learns, the founder considers whether to train her model in a supervised or unsupervised way, where supervised learning requires that the data have labeled outcomes and unsupervised learning does not. The cost of labeling outcomes on a broader dataset may be higher and the benefit of unsupervised learning on a smaller dataset may be lower, thus the type of learning of the model is likely to interact with the type of data chosen to produce different model results and different performances of the startup.

Finally, the founder considers who her startup should hire to build the AI model of cancer predictions, where the founder considers hiring either a full stack or a backend engineer. While it may be less obvious how the type of engineer hired interacts with the AI model feature selections to generate the startup's performance, the different skillsets of a full stack vs. a backend engineer are likely to generate different startup performances as function of model features selected. For example, a backend engineer will increase the startup's performance through her skills at building an accurate prediction model, especially with deep datasets and unsupervised learning which allow for the use of more cutting-edge techniques. However, a full stack engineer is likely to think critically about how the AI model feature choices will impact the end user experience, generating value for the startup using the simpler broad dataset and supervised learning approach which improve product usability.

Thus, in building an AI model to detect cancer, it will be important for the health tech startup founder to consider how the data type (breadth vs. depth), learning method (supervised vs. unsupervised), and hiring choice (full stack vs. backend engineer) selected interact to generate different performances for the startup. Figure 1 shows the strategic choice interdependencies for this startup founder building an AI model for cancer detection and how their combination impacts the overall performance of the startup.

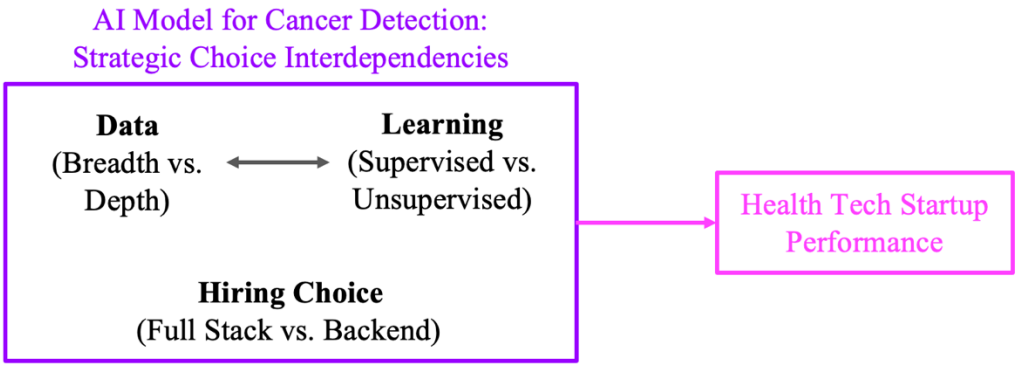
FIGURE 1
Strategic Environment of the Performance Links
Between the AI Model for Cancer Detection and the Health Tech Startup's Performance



To handle the complex links between the strategic choices around the AI model, the health tech startup's founder could use a mental model to guide her strategy selection. Consistent with previous research, her mental model could consist of associative performance links between a subset of the

choices, improving her decision-making by focusing her attention (Gavetti and Levinthal 2000, Csaszar and Levinthal 2016). For example, for the N strategic choices the founder needs to make (where N =3 for the data, learning, and hiring choices in this example), the startup founder may only consider the performance associations between a subset of M of these choices (where M = 2 for the data and learning choices in this example). Figure 2 shows this associative type of mental model, where the founder's mental model represents the performance dependence between the selection of data and learning type for her model while independently considering the choice of engineer to hire.

FIGURE 2
Mental Model with Associative Performance Links Selecting the AI Model Strategy

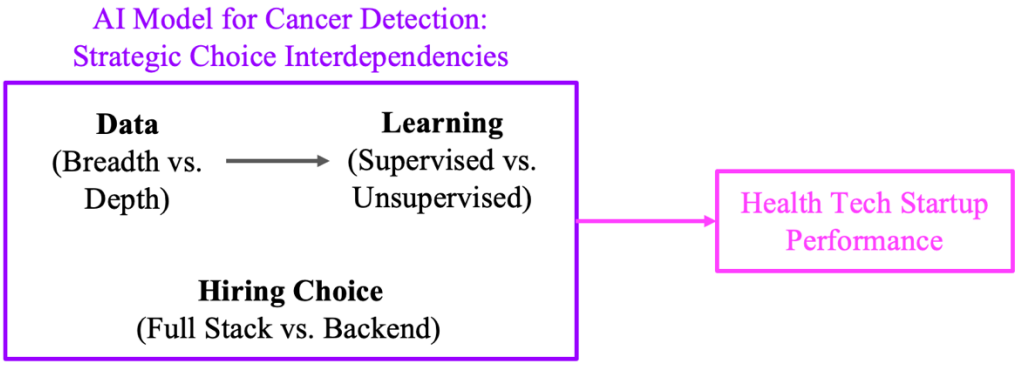


The representation of performance dependence between data and learning type as bi-directional in Figure 2 is important because research on mental accounting and strategic decision-making suggests that the way in which choices are represented impacts how those choices are evaluated and selected (see Thaler 1985, Kahneman and Tversky 2000). Particularly, the representation of data and learning type as a bundle of interdependent choices makes them likely to be evaluated and selected simultaneously (as in Thaler and Johnson 1990, Tripsas and Gavetti 2000). Thus, the start-up founder who holds the associative mental model in Figure 2 selects the data and learning type for her strategy jointly, while deciding on the choice of the engineer to hire to build the AI model independently.

Alternatively, the decision-maker could use a theory — a mental model with cause-and-effect performance links— to choose the AI model's features. As shown in Figure 3, this model also focuses on the links between a subset (M = 2) of the total (N=3) strategic choices: the data and the learning type.

However, unlike the associative model, the links in the theory have a causal direction, where the selection of the data type is seen as a performance antecedent of the learning type.

FIGURE 3
Mental Model with Cause-and-Effect Performance Links Selecting the AI Model Strategy



The ordering of strategic choices in the theory into an antecedent-consequent structure of performance dependence is likely to prompt a different strategy selection process than a bundle of strategic choices (see Camuffo et al. 2024, Denrell et al. 2004). Particularly, the antecedent-consequent logic proposes a natural sequential evaluation of choices, where an antecedent is chosen prior to its consequent. Thus, the theory in Figure 3 prompts the founder to first choose the data type (breadth vs. depth), and then choose the learning type of the model conditional on the data type selection. Like in the associative model, the hiring choice is selected independently.

The above example highlights that the way a decision-maker chooses a strategy depends on the performance link type (associative vs. cause-and-effect) in her mental model. Cause-and-effect performance links prompt sequential decisions, while associative performance links prompt simultaneous choice. Because these two types of mental models suggest different strategic decision-making processes, they are likely to lead the decision-maker to interpret performance feedback differently, resulting in different strategies and performance outcomes (Denrell et al. 2004, Martignoni et al. 2016). Next, I use a model to explore how mental models with associative and cause-and-effect performance links affect strategy selection and performance.

3. MODEL

Adapting the canonical NK-landscape model (Levinthal 1997, Gavetti and Levinthal 2000), I find that mental models with cause-and-effect performance links (or theories) outperform associative mental models in finding successful strategies because they help the decision-maker learn about the strategic environment through the efficient use of performance feedback. Table 1 below outlines the key parameters for the model.

TABLE 1: MODEL PARAMETERS

Object	**Symbol**	**Description**	**Category of Variance**
Time	T	The number of time periods, T = 100	System
Number of Simulations	S	The number of times to iterate through different landscapes, S = 10,000 unless noted	System
Number of Strategic Choices	N	Number of strategic choices in environment, N=10 unless noted	Strategic Environment
Interdependencies	K	Number of factors that the performance of each N choice is dependent on, K = 5 unless noted	Strategic Environment
Size of Mental Model (M)	M	The number of strategic choices in the mental model	Mental Model
Order	OR	A randomly selected order of length M of the form A→B→C→….. →M	Mental Model (for theory only)

3.1 Characterizing the Strategic Environment of Organizations as an NK-Landscape

Consistent with previous strategy research, I use the NK-landscape model to characterize the complex strategic environment of organizations (Levinthal 1997, see Baumann et al. 2019 for a review). In this model, strategy selection is conceptualized as a decision-maker traversing a 'rugged' landscape, where the difficulty of finding successful strategies depends on the number of strategic choices (N) and

the number of performance links between each strategic choice (K). The NK-landscape model presents a particular type of problem that faces decision-makers selecting strategies in organizational environments.

Often called an NP-hard or NP-complete problem in mathematics, the NK-landscape formalizes strategic environments that are so complex that past performance feedback cannot guide decision-makers to the best strategy (see strategy as an NP-hard problem: Siggelkow 2011, Ganco and Hoetker 2009, Hochba 1997). As these environments grow larger (high N) and more "rugged" (high K), it becomes harder to understand how each choice affects performance, since similar strategies can lead to very different outcomes (Rivkin 2000). Thus, decision-makers facing these complex strategic environments must find effective ways to use performance feedback to improve their strategic choices.

3.2 How Do Associative Mental Models and Theories Direct a Decision-Maker to Select Strategies and Use Performance Feedback?

The performance link type of a decision-maker's mental model determines how the decision-maker uses previous performance feedback to select future strategies (Denrell et al. 2004). Expanding on the intuition of the motivating example, I build a model of the simultaneous and sequential choice prompted by mental models with associative and cause-and-effect performance links. In this model, the health tech startup founder engages in ongoing strategic search rather than a one-time choice. She iteratively selects combinations of data type, learning type, and hiring over time, observing the resulting performance feedback, and updating future choices accordingly. Previously chosen strategies and their performance outcomes thus serve as inputs for subsequent strategic choices, with associative mental models and theories shaping how the founder uses this performance feedback.

In associative mental models the direction of the performance dependence between strategic choices is not specified, instead the model suggests a bundle of M strategic choices that are related. In our motivating example, the startup founder considers the data and learning type as a bundle of choices (M=2), which impacts both how the founder selects choices for her strategy and how she evaluates the performance feedback of those choices. Particularly, work in managerial cognition and routines shows

that when choices are grouped together, the performance implications of the choices are often considered jointly (Nelson and Winter 1982, Levinthal 2001, Tripsas and Gavetti 2000, Gavetti and Rivkin 2007).

As the founder collects performance feedback from prior strategies, for the M strategic choices in her mental model, her associative model prompts her to use this feedback to compare strategic choices in the bundle conditional on the values of all the other M choices in the mental model. This leads the decision-maker to use a relatively small quantity of prior performance feedback to evaluate any given combination of strategic choices in the mental model. It also means that no individual strategic choice in the mental model is treated as differentially important to strategic performance in the evaluation of prior performance feedback. For the N-M strategic choices not in the associative mental model, the startup founder will use the performance feedback to evaluate these choices independently and select the highest performing option.

To operationalize this bundling of choices in the model, at each time T, the decision-maker selects a random strategic choice from the set of M strategic choices in the mental model to change from its previous maximal value, for example, choice C. Then the decision-maker considers the previous performance of all unique combinations of the M strategic choices in the mental model which include the new value of choice C and selects the highest average performing combination. For the N-M strategic choices not included in the mental model she selects the highest performing value for each independently. If this process selects a strategy that has already been selected, the decision-maker randomly selects one of the N strategic choices to change. This approach is consistent with the operationalization of mental models in prior work (see Choi and Levinthal 2023, Csaszar and Levinthal 2016, Brusoni et al. 2007). At the end of time T when a strategy is selected (all N choices are made), the decision-maker receives performance feedback on the entire strategy.

In a theory, the performance dependence between strategic choices is directed, where the theory orders the performance dependence such that strategic choices are antecedents and consequents of one another. In the motivating example, the startup founder's theory suggests that the data type is a performance antecedent of the learning type (M is still equal to 2). This sequential strategic choice

process of a theory prompts the startup founder to use prior performance feedback differently, and ultimately more efficiently, to inform future strategic choice.

First, since the AI model's data type is only a performance antecedent in the theory, the decision-maker uses all previous performance feedback to select the best possible data type (for example a dataset with depth). Then, since learning type is a performance consequent of the data type choice, she uses prior performance feedback to select the best possible learning type conditional on the data type being depth (see Camuffo et al. 2024). As in the associative mental model case, she will use prior performance feedback to evaluate the N-M strategic choices not in the mental model independently.

To operationalize this sequential choice in the model, the decision-maker forms a theory where the performance links are randomly selected into an ordered linear chain of length M, i.e. A→B→…→M.[2] At each time T, the decision-maker selects a random strategic choice from the chain to change from its previous maximal value, for example, choice C. Then, using this chain of causal performance dependence and the newly selected value of C, the decision-maker, conditional on the selection of strategic choice C and its antecedents (A and B), selects the most performant value for strategic choice D. She continues this pattern of strategic choice using conditional performance dependence until she runs out of performance data, after which she explores with the next choice, and leaves the remaining M choices at their previous maximal value. Like with the associative models, cause-and-effect models also have the decision-maker decide on the N-M strategic choices not included in the model independently based on prior performance feedback, and modify a random N item if the strategy has been tried before. Once all strategic choices are made, the decision-maker receives performance feedback on the entire strategy (all N choices) at the end of each time T. Appendix A contains the pseudo-code and link to the full code for the operationalization of strategic search using both an associative mental model and a theory.

[2] This is a simple and common structure of cause-and-effect performance dependence in organizations, but there are other structures that are possible (see Pearl 2009, Georgakopoulos et al. 1995, Gary and Wood 2011).

Two important observations about the different use of performance feedback by decision-makers using theories vs. associative mental models follow from the operationalization above. First, theories represent a smaller number of nested conditional dependencies in their sequential strategic choice process, which leads the decision-maker to use a larger quantity of the prior performance feedback when selecting strategies than when using an associative mental model. One implication of this line of reasoning is that theories may be more cognitively efficient, because they prompt the decision-maker to consider fewer conditional dependencies overall. Second, theories assess the strategic environment efficiently to inform the chosen order, using more performance feedback for strategic choices that are earlier in the order of perceived strategic importance and less for those which are less primary.

Ultimately, the difference in the performance link type between associative mental models and theories leads to substantial differences in the way these mental models represent the strategic environment and direct decision-makers to evaluate performance feedback in the search for strategies over time. Table 2 summarizes these differences in representation and evaluation between associative mental models and theories. Because the causal structure of simple theories uses more performance feedback more efficiently to learn about strategic choice, theories should more accurately attribute performance in the strategic environment than associative mental models.

TABLE 2: DIFFERENCES IN REPRESENTATION AND EVALUATION BETWEEN ASSOCIATIVE MENTAL MODELS AND THEORIES

	Associative Mental Model	**Theory**
	Difference in Representation	
Type of Performance Link	Associative, Undirected: A←→B	Causal, Directed: A→B
Number of Performance Links[3]	Maximal: $M * (M - 1)$	Minimal: $\frac{M*(M-1)}{2}$
	Difference in Evaluation	
Evaluation Structure	Simultaneous: Mutual conditional evaluation across all M Choices	Sequential: Later choices evaluated conditional on earlier choices only
Data Use per Strategic Choice	Less: Data dispersed across each unique combination of M choices	More: Data aggregated by causal antecedents only
Ordering of Strategic Importance	None: [A, B, C, … M]	Hierarchical: A→B→C→….. →M
Data-Importance Alignment	Equal: Same data use and importance for all M strategic choices	Greater: Use more data to make more important strategic choices

3.3 How Accurately Do Associative Mental Models and Theories Attribute Performance?

To show that the differential use of performance feedback leads theories to better learn about performance in the strategic environment, I model two measures of performance attribution: maximum performance attribution and average performance attribution. These two measures capture the joint difficulty of searching for strategies, especially in the NK literature, which is often discussed in the trade-off between exploration and exploitation (March 1991, Siggelkow and Rivkin 2006). A decision-maker

[3] See Appendix C for explanation and calculation of the number of performance links in associative mental models and theories as a function of the number of strategic choices in the mental model (M).

searching for a strategy has two concerns: to find the best possible strategy in the environment (called the global maximum) and to improve average performance over time. In general, these goals are presented as a trade-off, where a decision-maker must forgo average performance improvements to explore, or forgo learning about the best possible strategies in the environment to exploit. However, the more efficient use of performance feedback through the sequential choice process of theories allows a decision-maker to better learn about both the best strategy and better strategies on average in the environment.

To measure performance attribution, at the end of the T time periods of search, I analyze the strategies tried and performance received by the decision-maker, which indicates how well the decision-maker has learned what drives maximal and average performance in the environment through the search process. For the maximum performance attribution measure, I take all T strategies tried and their respective performance, and average performance for each strategic choice independently, selecting the strategic value for each N strategic choice with the highest performance to generate a predicted maximum strategy based on performance feedback. Then I compare the value of each N strategic choice from the decision-maker's search to the values of strategic choice at the actual global maximum in the strategic environment, and divide the number of strategic choices from the predicted maximum that are equal to the value of the global maximum in the environment by the total number of choices in the environment N. Equation 1 formalizes this proportion of global maximum values correctly identified by the decision-maker as the maximum performance attribution measure. Figure 4 visualizes the maximum performance attribution based on strategic search using associative mental models and theories across varying sizes of mental models (M) (for N=10, K=5).

$$Maximum\ Performace\ Attribution = \frac{\sum_{j=1}^{N} if(max_j\left(\frac{\sum_{i=0}^{T} Performance_{f(j)=0}}{T}, \frac{\sum_{i=0}^{T} Performance_{f(j)=1}}{T}\right) = \mathrm{K}_j\ ,1,0)}{N} \quad (1)$$

where K_j is the global maximum value for strategic choice j in the environment.

FIGURE 4

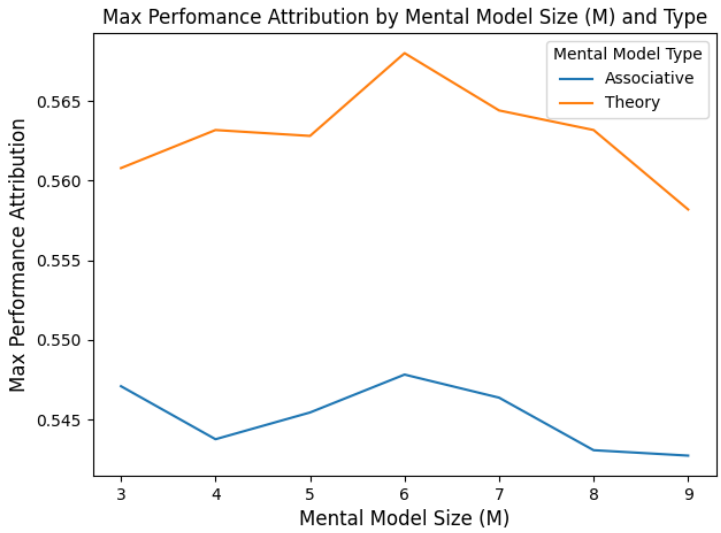


Figure 4 shows that the proportion of strategic choices that are correctly attributed as maximally performant is significantly higher for strategies selected using theories than for those selected using associative mental models. Particularly, decision-makers using theories to select strategies attribute performance to global maximum values approximately 2% more than associative mental models. This means that the way that theories use prior performance feedback to sample strategies improves the decision-maker's ability to identify maximally performant strategies in the environment better than associative mental models over time. However, decision-makers care both about maximal and average performance of strategies, and often these outcomes conflict with each other in searching for strategies across NK-landscapes.

To measure average performance attribution, I correlate the average performance difference between the two values for each N strategic choice from the decision-maker's strategic performance feedback with the average performance difference between the two values for each N strategic choice from the environment. This quantifies how strongly related the decision-maker's understanding of how each N strategic choice determines average performance is to what actually drives average performance in the environment. Equation 2 formalizes this correlation between the average performance of choices in the decision-maker's performance feedback and the average performance of choices in the strategic

environment as average performance attribution. Figure 5 visualizes average performance attribution based on strategic search using associative mental models and theories across varying sizes of mental models (M) (for N=10, K=5).

$$Average\ Performance\ Attribution = Cor([\frac{\sum_{i=0}^{T} Performance_{f(j)=1}}{T} - \frac{\sum_{i=0}^{T} Performance_{f(j)=0}}{T}]_{j=1}^{N}, \frac{\sum_{l=0}^{N*N} Performance_{f(k)=1}}{T} - \frac{\sum_{l=0}^{N*N} Performance_{f(k)=0}}{T}]_{k=1}^{N}) \quad (2)$$

where there are N*N possible strategies in the environment each with a performance value.

FIGURE 5

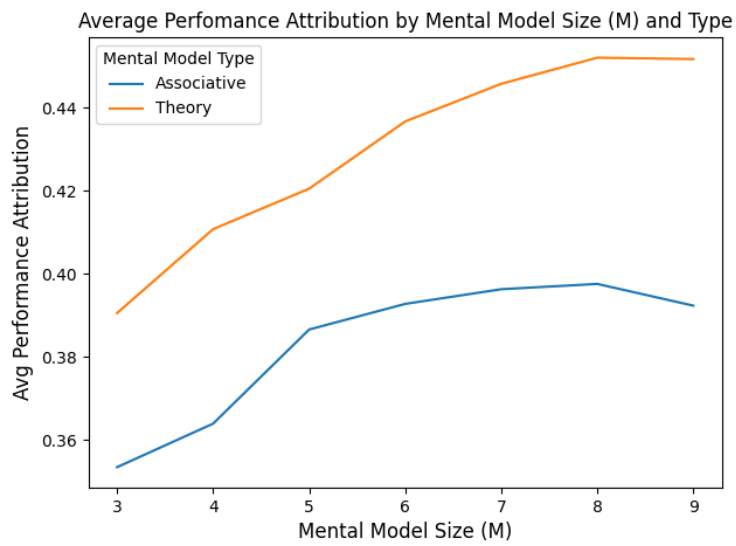


Figure 5 shows that decision-makers using a theory to select strategies better attribute average performance in the environment than those using an associative mental model, where theories select strategies with between a 2 -6% higher average correlation between strategic choices and performance. This means that a decision-maker using a theory successfully navigates what is often considered a trade-off in searching for strategies on an NK-landscape. Particularly, the simple structure of a theory leads to a more efficient use of prior performance feedback to select future strategies, which both improves the decision-maker's understanding of maximally performant strategies and average performing strategies relative to associative mental models. Since decision-makers using a theory to select strategies better understand what drives both maximum and average performance in the environment, they should select different and ultimately better strategies than those using an associative mental model.

3.4 Do Decision-Makers Using Theories Select Different and Better Strategies Than Those Using Associative Mental Models?

Even though decision-makers in the model start with the same initial strategy in the same strategic environment and use the same M strategic choices in their mental models, the difference in how theories and associative mental models use performance feedback to attribute performance leads theories to select significantly different and better strategies over time. Figure 6 below shows the average proportion of shared strategies selected by a decision-maker using the same M strategic choices in an associative mental model vs. theory across 100 times periods and 10,000 simulation runs (for strategic environment, N = 10, K = 5).

FIGURE 6

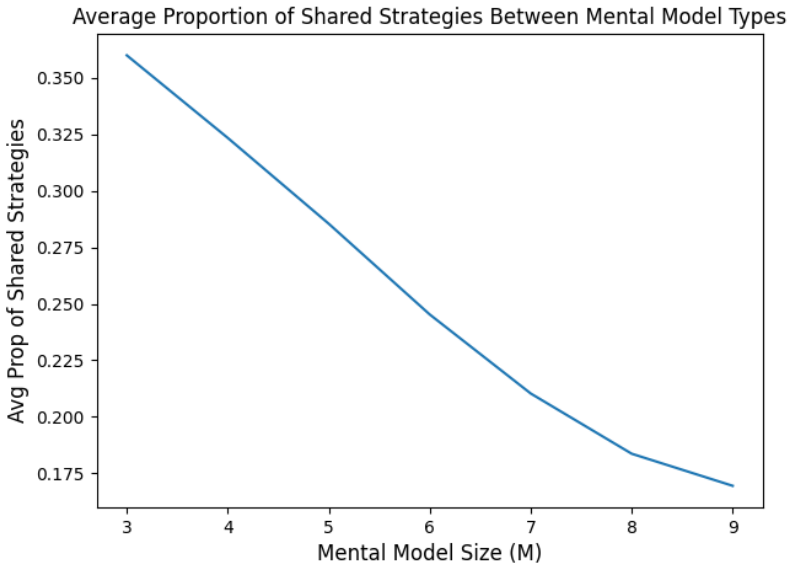


Figure 6 shows that the difference in link type of the mental models leads the decision-maker to select different strategies between 65 to 85 out of 100 times, depending on the size of the representation used. The only source of variation between the strategic search processes of associative mental models and theories is the performance link type between the M strategic choices considered. Thus, I conclude that the respective simultaneous and sequential selection and evaluation of strategic choice of associative mental models and theories generates a significant difference in the decision-maker's search for strategies.

To model the performance consequences of the different strategies selected using an associative mental model vs. theory, I run 10,000 iterations for 100 times periods of mental models of varying size M (from 3 to 9), where both associative mental models and theories search for strategies across NK-landscapes as described above. Figure 7 shows the performance of strategies found by decision-makers using associative mental model vs. theory of size M (with N = 10 and K = 5). These results are stable across strategic environments of varying N and K, as shown in Appendix B.

FIGURE 7

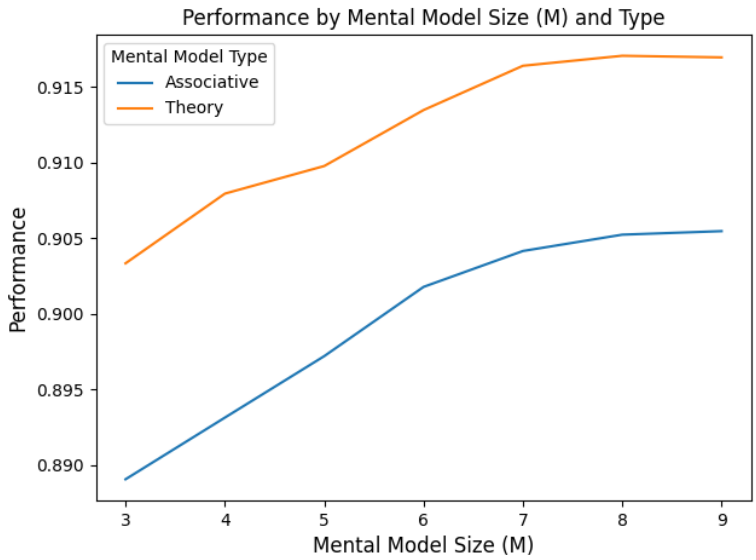


Figure 7 shows two important results. First, mental models that contain more strategic choices (high M) perform better than those that contain fewer (low M). This is line with previous research that finds when the dimensions in the strategic environment do not vary in their importance for determining performance, maximizing the number of strategic choices in the representation is beneficial (Csaszar and Levinthal 2016). Second, for a mental model at any given size M, theories outperform associative mental models.

The finding that a simpler mental model (theory) outperforms a more complicated associative mental model is in line with the previous research that suggests that one of the ways that mental models improve strategic search is by simplifying complex strategic environments (Gavetti and Levinthal 2000). However, this finding provides two novel contributions to the literature. First, prior work mostly

considers how mental models simplify the number of strategic choices in the environment (Csaszar and Levinthal 2016, Gavetti and Levinthal 2000, see Martignoni et al. 2016 for an exception). The results above add to the mental models and theory-based view literature by suggesting that theories simplify the strategic environment and improve performance by reducing the number of interdependencies considered by decision-makers, not the number of strategic choices.

Second, while previous work suggests that simple mental models can improve strategic search, it also shows that accurate mental models are likely to generate better strategies (Gary and Wood, 2011). While theories remain simpler representations than associative mental models across all strategic environments, as environments become more interdependent (higher K relative to N), theories are less accurate representations. Thus, my results, which show that theories are, on average, more performant than associative mental models, suggest that simpler representations of interdependencies outperform more accurate representations of interdependencies in highly interdependent environments. In the next section, I explore this trade-off between simplicity and accuracy in representation of interdependencies to connect my results that theories outperform associative mental models to the core explanations of why mental models improve strategic search in the prior literature.

3.5 Connecting the Better Performance of Theories to their Ability to Accurately and Simply Represent the Strategic Environment

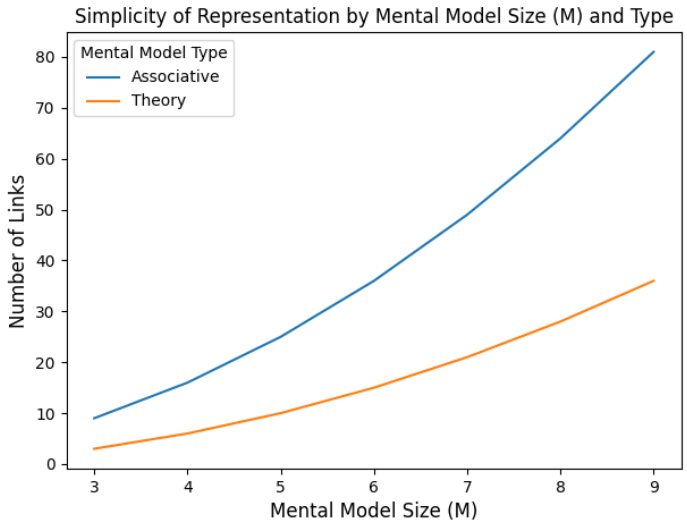
Prior literature has traditionally highlighted two features of mental models that help decision-makers search for more performant strategies: simplicity and accuracy. Simple mental models help decision-makers find better strategies by shrinking and smoothing the space the decision-maker searches over, which generates a faster and more efficient search process (Gavetti and Levinthal 2000). While accurate mental models help decision-makers by correctly characterizing the performance variation in the environment, meaning decision-makers are better able to use performance feedback to locate better strategies (Gary and Wood 2011).

The ability for a mental model to simply and accurately represent the strategic environment varies as a function of the number of strategic choices (N) and the number of performance link dependencies (K)

in the environment. In simple environments (low N and low K), mental models can be both accurate and simple. But as complexity grows (higher N and higher K), mental models face a trade-off between simply and accurately representing the environment.

To model how simply a mental model represents the strategic environment, I calculate how many performance links a mental model of M strategic choices will consider. In general, associative models include all possible performance links between choices since they suggest a performance relationship between choices, but don't specify a causal direction to the performance dependency between them. In contrast, for theories which represent chains of cause-and-effect links, the decision-maker implicitly considers the performance consequences of all preceding strategic choices in the mental model. Appendix C includes a set of formal equations and a proof that, as defined in this model, theories will always represent fewer performance dependencies than associative mental models. Figure 8 visualizes the number of performance dependencies considered in associative mental models and theories across varying sizes of mental models M (for N=10, K=5).

FIGURE 8



Because the number of performance links represented in each mental model does not depend on the number of strategic choices (N) or the number of actual performance links between these choices (K), this relationship will hold across all strategic environments. Thus, for a given set of M strategic choices, cause-and-effect models will always be simpler representations than associative models. However, while

the simplicity of a representation does not depend on the environment's complexity, the accuracy of each model does.

To model how accurately a mental model represents the strategic environment, I measure the degree to which its performance links match the true performance links in the environment. Across each NK-landscape, I compare how accurately the generated mental models of size M— the mental model with associative performance links vs. the theory with a *randomly selected set* of cause-and-effect performance links—represent the true performance dependencies in the strategic environment. Figures 9 and 10 show these results: Figure 9 compares model link accuracy as the number of strategic choices (N) increases (with K = 5), and Figure 10 compares link accuracy as the number of performance links (K) increases (with N = 10). Appendix D provides a mathematical in expectation proof of these results.

FIGURE 9: Accuracy of Links by Mental Model Size (M) and Type Across Number of Strategic Choices (N)

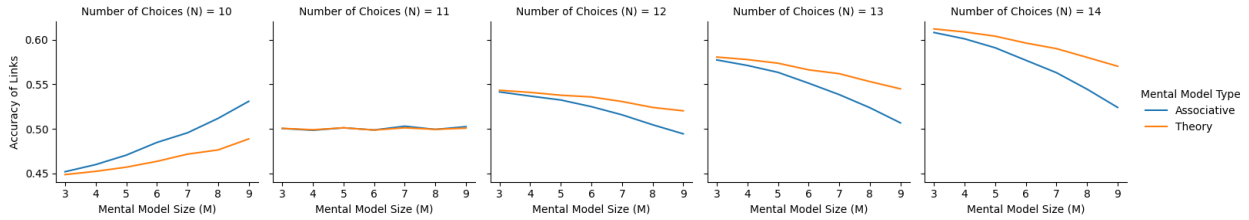


FIGURE 10: Accuracy of Links by Mental Model Size (M) and Type, Across Interdependence of Strategic Choices (K)

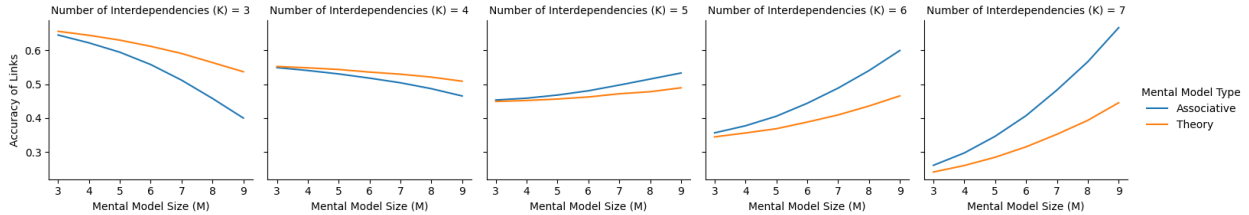


Figure 9 above show that as the number of strategic choices (N) increases, cause-and-effect models are more accurate across all mental model sizes (M). Figure 10 shows when the number of links between choices (K) increases (with N fixed at 10), the cause-and-effect models are more accurate when K is below 5 and less accurate when K is above 5. The expected accuracy of associative mental models vs. theories in strategic environments can be formalized as a function of N and K (see equation D2 in Appendix D), where theories are more accurate in expectation in environments that are less densely connected (relatively high N and relatively low K).

The prior literature suggests that mental models that are accurate and simple improve performance in the search for strategies (Gavetti and Levinthal 2000, Gary and Wood 2011). The section above shows that when the strategic environment is relatively simple (low N, low K), mental models can be both accurate and simple, and in these cases, theories are more accurate and simpler representations than associative models. As the number of strategic choices increases (N), theories remain both simpler and more accurate representations. But when complexity comes from a greater number of performance links between choices (K), theories are simpler, while associative mental models are more accurate.

Thus, in explaining why theories are more performant than associative models, for most strategic environments, the explanation of the prior literature remains consistent. Theories are simpler and more accurate representations of the strategic environment at any number of strategic choices (N) and at relatively low interdependence between choices (K), and this simplicity and accuracy of representation improves strategic search. However, for environments with relatively high interdependence (K) compared to the number of strategic choices (N), theories are simpler representations and associative mental models are more accurate.

The results in the above sections are run in this type of highly interdependent strategic environment (N = 10, K = 5) where there is a trade-off between accurate associative mental models and simple theories, showing that the simplicity of theories outperforms the accuracy of associative mental models in the selection of strategies. In these highly interdependent strategic environments, while theories are less accurate, their simplicity allows decision-makers to more accurately learn about the strategic environment (see performance attribution results of Figure 4 and 5), which is why the simplicity of theories ultimately outperforms the accuracy of associative mental models.

4. DISCUSSION AND CONCLUSION

The cause-and-effect performance links in theories help decision-makers choose successful strategies through improving their understanding of what drives strategic performance. The simple, sequential representation of strategic choices by theories improves the quantity and quality of

performance feedback used by decision-makers, leading to better performance attribution and ultimately the selection of better strategies over time when compared to associative mental models. This is true even in highly interdependent environments where there is a trade-off between the accuracy of associative mental models and the simplicity of theories, showing that theories are a more parsimonious mental model for selecting successful strategies.

Overall, the above results, which suggest that theories are, on average, better than associative mental models at finding successful strategies in interdependent strategic environments, prompt a series of questions and, hopefully, a series of paths for future research. In the sections below I ask several of these questions and suggest some promising areas for addressing them through further work.

4.1 Simplicity Over Accuracy?: Simple, Causal Mental Models Provide Structure that Helps Incorporate Performance Feedback

A key question these results raise is: why does simplicity outperform accuracy in strategic environments that require a trade-off between the two? Research on mental models in strategy, and on human cognition more generally, shows that people often use simple models to make effective decisions in complex situations (Bingham and Eisenhardt 2011, Schwenk 1984, Simon 1977, Gigerenzer, 2008). However, there is a debate over whether causal reasoning actually improves decision-making (Sperber et al. 1995, Mercier and Sperber 2017).

My findings suggest that cause-and-effect links make mental models simpler in a way that helps decision-makers better use performance feedback in complex environments. In these settings, the main challenge is turning past feedback into better future strategies (Siggelkow 2011, March 1991). When there is no true solution to environmental complexity (strategy as an NP-hard problem), mental models with cause-and-effect performance links provide the next best thing — a simple structure of sequential choices that makes performance feedback easier to understand (see Pearl 2009, Denrell et al. 2004).

This result can be seen as a formalization of the empirical findings in the theory-based view of strategy, where holding a theory improves experimentation and allows for the application of the scientific

method to organizational strategy (Camuffo et al. 2024, Camuffo et al. 2020). I outline two implications of the simplicity of theories outperforming more accurate associative mental models below.

First, my results suggest that evaluations of the quality of a theory should be separated from evaluations of the performance benefits of holding one. While previous work shows that iterating with theories can improve the accuracy of performance understandings (Camuffo et al. 2024, Ehrig and Schmidt 2022), my work shows that performance gains may come from how the theory structures decision-making and performance evaluation, which does not depend on how accurate the theory is. Future research should separate these two types of benefits.

Second, I propose that theories lead decision-makers to choose strategies in a step-by-step (sequential) way, while associative mental models lead to all-at-once (simultaneous) decisions. Beyond validating this basic difference, future research should explore how these mental models create distinct decision-making processes, especially in how strategic choices are ordered.

4.2 Does it Really Not Matter if My Theory is "Good"?

Because my model compares the average performance of theories versus associative mental models, it doesn't show how individual theories—each with different levels of accuracy—perform. This raises another question: does a theory's performance depend on how accurate, or "good", it is?

While my results show that accuracy of representation doesn't always generate performance, they cannot directly answer this question because of how the baseline version of the NK-landscape works. In this model, it is assumed that every strategic choice (N) depends on K others, and while some choices have more impact than others through the weight matrix (W), because of this equal-K dependence, it is difficult to conceptualize any strategic choice as more or less causally central to the performance of a strategy. Because the strategic environment is largely causally non-ordered, most theories which causally order strategic choices are equally as accurate or "good." Future research should explore how theories affect decision-making in more structured (or "ordered") environments (see Csaszar and Levinthal 2016 use of parameter W for an example) —where some theories clearly reflect the environment better than others and might perform differently as a result.

4.3 What is the Role of Theories in Generating Group Consensus?

Recent research on strategic decision-making in organizations has emphasized the need for group consensus (Adner and Levinthal 2024, Carroll and Sørensen 2021). My findings show that theories help individuals navigate complex environments, but they may also lead to more disagreement through what I call the "chicken-and-egg effect". While everyone may agree two things are related, theories which assign a clear causal direction (e.g., chicken before egg or vice versa) are more likely to spark conflict. Future research should explore how assigning causal direction can create disagreement in groups and look for ways to reduce this downside of using theories in strategic decision-making.

4.4. Conclusion: Any Old Theory Will Do?

An old poem adopted by organizational scholars told the story of lost soldiers who found their way back to camp using a map of the wrong mountain range—and from it came the phrase "any old map will do" (Holub 1977). Similarly, my results show that even a randomly generated theory can outperform a more accurate but complex associative model in hard-to-learn environments. This suggests that having a simple theory to guide strategy may be like having the wrong map: helpful, even though it is inaccurate.

Just as water flows towards valleys in a mountain range, complex strategic environments still have patterns linking choices to performance. A mental model that assumes a clear (even if incorrect) path through the environment may better detect these patterns than one that tries to track everything (see Denrell et al. 2004). This helps explain why simpler theories can outperform more accurate models when the environment is too complex to fully understand—a finding supported by broader research on heuristics and cognitive shortcuts (Bingham and Eisenhardt 2011, Simon 1977, Gigerenzer 2008). Overall, my model shows that theories find more successful strategies because they offer a simpler way to navigate and understand complexity.

REFERENCES

Adner R, Levinthal DA (2024) Strategy Experiments in Nonexperimental Settings: Challenges of Theory, Inference, and Persuasion in Business Strategy. *Strategy Science*.

Baumann O, Schmidt J, Stieglitz N (2019) Effective Search in Rugged Performance Landscapes: A Review and Outlook. *Journal of Management* 45(1):285–318.

Bingham CB, Eisenhardt KM (2011) Rational heuristics: the 'simple rules' that strategists learn from process experience. *Strategic Management Journal* 32(13):1437–1464.

Brusoni S, Marengo L, Prencipe A, Valente M (2007) The value and costs of modularity: a problem-solving perspective. *European Management Review* 4(2):121–132.

Camuffo A, Cordova A, Gambardella A, Spina C (2020) A Scientific Approach to Entrepreneurial Decision Making: Evidence from a Randomized Control Trial. *Management Science* 66(2):564–586.

Camuffo A, Gambardella A, Pignataro A (2024) Theory-Driven Strategic Management Decisions. *Strategy Science* 9(4):382–396.

Carroll GR, Sørensen JB (2021) *Making Great Strategy: Arguing for Organizational Advantage* (Columbia Business School Publishing, New York City).

Choi J, Levinthal D (2023) Wisdom in the Wild: Generalization and Adaptive Dynamics. *Organization Science* 34(3):1073–1089.

Csaszar FA, Levinthal DA (2016) Mental representation and the discovery of new strategies. *Strategic Management Journal* 37(10):2031–2049.

Csaszar FA, Ostler J (2020) A Contingency Theory of Representational Complexity in Organizations. *Organization Science* 31(5):1198–1219.

Denrell Jerker, Fang Christina, Levinthal DA (2004) From T-Mazes to Labyrinths: Learning from Model-Based Feedback. *Management Science* 50(10):1366–1378.

Ehrig T, Schmidt J (2022) Theory-based learning and experimentation: How strategists can systematically generate knowledge at the edge between the known and the unknown. *Strategic Management Journal* 43(7):1287–1318.

Felin T, Gambardella A, Zenger T (2024) Theory-Based Decisions: Foundations and Introduction. *Strategy Science* 9(4):297–310.

Ganco M, Hoetker G (2009) NK modeling methodology in the strategy literature: Bounded search on a rugged landscape. *Research Methodology in Strategy and Management* (Emerald Group Publishing Limited), 237–268.

Gary MS, Wood RE (2011) Mental models, decision rules, and performance heterogeneity. *Strategic Management Journal* 32(6):569–594.

Gavetti G, Levinthal D (2000) Looking Forward and Looking Backward: Cognitive and Experiential Search. *Administrative Science Quarterly* 45(1):113–137.

Gavetti G, Rivkin JW (2007) On the Origin of Strategy: Action and Cognition over Time. *Organization Science*.

Georgakopoulos D, Hornick M, Sheth A (1995) An overview of workflow management: From process modeling to workflow automation infrastructure. *Distrib Parallel Databases* 3(2):119–153.

Gigerenzer G (2008) Why Heuristics Work. *Perspect Psychol Sci* 3(1):20–29.

Hochba DS (1997) Approximation Algorithms for NP-Hard Problems. *SIGACT News* 28(2):40–52.

Holub M (2014) Brief Thoughts on Maps. *A Geological Miscellany* (Princeton University Press), 3–4.

Kahneman D, Tversky A eds. (2000) *Choices, Values, and Frames* (Cambridge University Press, New York : Cambridge, UK).

Kaplan S (2011) Research in Cognition and Strategy: Reflections on Two Decades of Progress and a Look to the Future. *Journal of Management Studies* 48(3):665–695.

Levinthal DA (1997) Adaptation on Rugged Landscapes. *Management Science* 43(7):934–950.

Levinthal DA (2001) Organizational Capabilities in Complex Worlds. Dosi G, Nelson RR, Winter S, eds. *The Nature and Dynamics of Organizational Capabilities* (Oxford University Press), 0.

March JG (1991) Exploration and Exploitation in Organizational Learning. *Organization Science* 2(1):71–87.

Martignoni D, Menon A, Siggelkow N (2016) Consequences of misspecified mental models: Contrasting effects and the role of cognitive fit. *Strategic Management Journal* 37(13):2545–2568.
Mercier H, Sperber D (2017) *The enigma of reason* (Harvard University Press, Cambridge, MA, US).
Nelson RR, Winter SG (1985) *An Evolutionary Theory of Economic Change* (Harvard University Press).
Pearl J (2009) *Causality* 2nd ed. (Cambridge University Press, Cambridge).
Porac JF, Thomas H, Baden-Fuller C (1989) Competitive Groups as Cognitive Communities: The Case of Scottish Knitwear Manufacturers. *Journal of Management Studies* 26(4):397–416.
Rivkin JW (2000) Imitation of Complex Strategies. *Management Science* 46(6):824–844.
Schwenk CR (1984) Cognitive simplification processes in strategic decision-making. *Strategic Management Journal* 5(2):111–128.
Siggelkow N (2011) Firms as Systems of Interdependent Choices. *Journal of Management Studies* 48(5):1126–1140.
Siggelkow N, Rivkin JW (2006) When exploration backfires: Unintended consequences of multilevel organizational search. *Academy of Management Journal* 49(4):779–795.
Simon HA (1977) The Logic of Heuristic Decision Making. *Models of Discovery* (Springer, Dordrecht), 154–175.
Sorenson O (2024) Theory, Search, and Learning. *Strategy Science* 9(4):372–381.
Sperber D, Premack D, Premack AJ (1995) *Causal cognition: A multidisciplinary debate* (Clarendon Press/Oxford University Press, New York, NY, US).
Thaler R (1985) Mental Accounting and Consumer Choice. *Marketing Science* 4(3):199–214.
Thaler RH, Johnson EJ (1990) Gambling with the House Money and Trying to Break Even: The Effects of Prior Outcomes on Risky Choice. *Management Science* 36(6):643–660.
Tripsas M, Gavetti G (2000) Capabilities, cognition, and inertia: evidence from digital imaging. *Strategic Management Journal* 21(10–11):1147–1161.
Valentine J, Novelli E, Agarwal R (2024) The Theory-Based View and Strategic Pivots: The Effects of Theorization and Experimentation on the Type and Nature of Pivots. *Strategy Science* 9(4):433–460.